

Lab 1: Requirements Engineering & UML Use-Case Modelling

USE-CASE FLOW SPECIFICATION

Software Engineering

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Use Case: Match & Allocate Supplies

Preconditions

1. Disaster Manager is logged into the system.
2. A shelter supply request has been submitted.
3. The request contains the required supply type and quantity.
4. Current warehouse inventory information is available.

Main Success Scenario

1. Disaster Manager selects a pending shelter supply request.
2. System displays the requested supply type, quantity, and priority.
3. System checks the available warehouse inventory.
4. System identifies matching available supplies.
5. System determines the delivery priority.
6. System allocates the required supplies to the request.
7. System updates the available warehouse quantity.
8. System records the allocation and prepares the request for delivery.
9. System displays an allocation confirmation.

Postconditions

1. Required supplies are allocated to the shelter request.
2. Warehouse inventory is updated.
3. Delivery priority is recorded.
4. The request is ready for logistics/delivery processing.

Alternate Flow – Insufficient Supply Available

1. System checks the warehouse inventory.
2. The requested quantity is not fully available.
3. System displays the available quantity.
4. System partially allocates the available supplies.
5. System marks the remaining quantity as pending.
6. System notifies the Disaster Manager about the shortage.